AI Fashion Design Studio

Project Team

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Session 2021-2025

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June, 2025

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Chapter 1

Introduction

This project proposal aims to provide a clear and detailed overview of the AI Fashion Design Studio project. The document outlines the key objectives, features, and overall vision for the system. Its purpose is to guide everyone involved like developers, designers, stakeholders, and potential investors on what the project is about, how it will work, and what it hopes to achieve.

1.1 Problem Statement

Fashion design has been plagued by a long development cycle, difficulty in imagining the final effect, and many shortcomings in customization and selling. These hurdles not only stifle creativity and increase production costs but also impede the ability to meet market demands for personalized fashion. AI Fashion Design Studio seeks to change this status quo, and greatly improve each step in the traditional fashion design process through generative algorithms. Our platform expedites the initial design phase through the Template Management module, which allows designers to quickly access and customize a broad array of fashion templates. The Sketch to Image Conversion module transforms rudimentary hand-drawn sketches into detailed digital designs, significantly cutting down the time required to transition from concept to prototype. Further refining the design process, our 3D Visualization tool allows designers to inspect detailed 3D models of their creations from any angle, thus eliminating the guesswork from product visualization.Virtual try-on module provides the custom fitting experience wherein instant feedback on fit and style can be retrieved-an important step forward in helping develop personalization and reduce return rates in e-commerce. To support the commercialization of the finalized products, our system includes dedicated modules for creating dynamic marketing content whereby the Clothing Image Ads and Video Ads modules automatically generate high-quality promotional materials that showcase the garments in various poses and scenarios.

By streamlining these critical phases of the fashion design and marketing workflow, AI Fashion Design Studio not only enhances efficiency and reduces time-to-market but also revolutionizes how fashion products are designed, presented, and sold, offering an innovative solution to the age-old challenges of the fashion industry.

1.2 Scope

The AI Fashion Design Studio was developed to make the transition from initial design to a market-ready products, focusing only on adult clothing; this studio does not support accessories, makeup, children's clothing, or any items not classified as clothing. The important functions in the project include converting initial sketches into digital images, building highly detailed 3D models for visualization purposes, and enabling virtual try-on fitting for assessing fit and style. However, it does not allow for real-time 3D editing and modifications: it does set pre-adjustments automatically, emphasizing setting benchmarks and having automatic processing. Marketing content generation is also automated, producing both image and video advertisements to facilitate the promotion of finished designs. It is a web application system, based on a tech stack that includes PyTorch, MERN(MongoDB, Express.js, React, Node.js) for full-stack development, FastAPI for high-performance backend services, and Hugging Face for leveraging state-of-the-art generative models.

1.3 Modules

1.3.1 Module 1 : Template Management

This module manages access to a library of pre-designed templates and provides a database of ready-made clothing design templates for user selection.

1.3.2 Module 2: GUI Development

This module will develop the graphical user interface for the application.

1. We will design and implement the front-end interface where users interact with the system.

2. We will integrate the interface with back-end processes to ensure seamless user experience.

1.3.3 Module 3: Sketch to Image Conversion

Utilizing a fine-tuned LoRa version of Stable Diffusion, this module transforms hand-drawn sketches into high-quality 2D digital images. It combines delightful and creative user prompts that can include sketches on every developed detail and a highly accurate digital picturing.

1. User uploads sketch and textual description of the desired output.
2. Convert user-uploaded sketches into digital images using fine-tuned SDXL LoRA model.

1.3.4 Module 4: Virtual Try-On + 3D Visualization

This module will enable virtual try-on experiences and create detailed 3D models of clothing designs for comprehensive review.

1. Enables users to virtually try on and interact with clothing designs using digital avatars.
2. Converts 2D virtual try-on into 3D virtual try-on that can be viewed and manipulated in a 360-degree environment.

1.3.5 Module 5: Crossover Design Mechanism

This module will generate unique new designs by combining elements from existing garments

1. Allow users to select two existing designs for combination.
2. Automatically generate a new design that incorporates elements of the chosen garments.

1.3.6 Module 6: Marketing Material Generation

This module automatically generates marketing materials, leveraging diffusion models to create visually appealing image and video ads.

1. Generate clothing image ads (3-5 images) featuring model in various poses wearing the designed clothes.
2. Create short 5-10 second video ads showcasing the clothes in dynamic, custom scenes.

1.4 User Classes and Characteristics

User Class	Description
Fashion Designers	Fashion designers are the primary customers of this system because they directly use AI Fashion Design Studio to improve and optimise the design process. They interact with the system from the sketch all the way to the final and polished version of the product.Designers take advantage of modules including the Template Management for base designs, turning Sketch to Image Conversion is used for digital rendering and 3D Modeling concerning more detailed visualization. The system fulfills their need for rapid prototyping and allows for high customization with tools that support creative freedom and precision.
E-commerce Retailers	E-commerce retailers use the platform to create appealing online catalogs and provide their customers with an innovative shopping experience. T They mostly use the Virtual Try-On and Marketing Content Creation modules (Clothing Image Ads and Video Ads) to produce realistic presentation and advertising contents which would be expected to have higher tendency of attracting the attention of buyers and thus boost sales. The platform helps them reduce return rates by allowing customers to visualize products accurately before purchasing.
Customers	End consumers benefit from the Virtual Try-On and Crossover Design Mechanism modules, which provide a personalized shopping experience. They can see how garments look on avatars that match their body type and even participate in designing custom wear by selecting elements they like, meeting the demand for personalized fashion.

Table 1.1: User Classes and Characteristics

Chapter 2

Project Requirements

2.1 Use-case

The AI Fashion Design Studio offers an intuitive workflow diagram that visually represents how users interact with the system from inception to final outputs. The process initiates when the user either uploads a sketch or selects from a library of templates. These actions trigger two paths: the Sketch to Cloth Image conversion for sketches and direct engagement with the Virtual Try-On for templates. Both paths converge at the Virtual Try-On phase where designs are fitted on digital avatars for accurate representation. Subsequently, 3D Visualization offers an enhanced, detailed view of the outfits. The process culminates in the creation of marketing content, where the system automatically generates tailored image ads and video ads for showcasing the designs in various scenarios. This diagram clearly illustrates all possible interactions a user can have with the system, allowing for seamless navigation through each phase of the fashion design process.

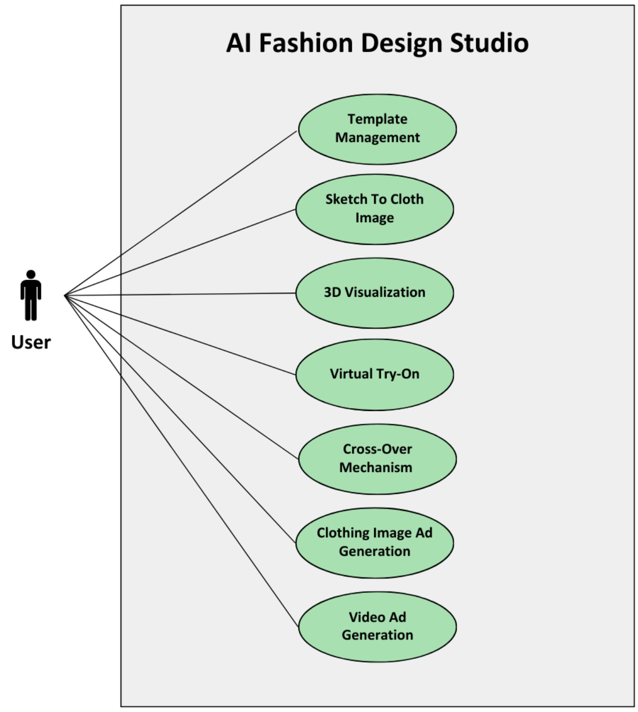

Figure 2.1: Use Case Diagram

Table 2.1: Use-Case Table

User Class	Actor	Description	System Response
Upload Sketch/Image	User	User uploads a sketch or an image of a garment.	System processes the upload and prepares it for transformations.
Select Template	User	User selects a template from a database.	System retrieves the selected template.
Sketch to Image Conversion	User	Converts uploaded sketch to a detailed 2D image using AI.	System displays the converted image for review.
Crossover Design Generation	User	User combines two designs to initiate a crossover design.	System merges designs and displays the new design.
Virtual Try-On	User	User selects a garment for virtual try-on on an avatar.	System fits the garment on the avatar.
3D Visualization	User	User requests a 3D visualization of the garment.	System generates a 3D model and displays it.
Generate Clothing Image Ads	System	Generates clothing image ads from finalized designs.	System creates and stores ads in the ad database.
Create Video Ads	System	Produces video ads featuring the clothes in dynamic scenes.	System generates and stores video ads for marketing.

2.2 Functional Requirements

This section provides a list of functionalities of each of the proposed modules of the AI Fashion Design Studio, explaining users' engagements and predicted outcomes in a software engineering perspective:

2.2.1 Module 1: Template Management

Following are the requirements for module 1:

1. Functionality: Users can browse and select from a provided list of fashion templates.
2. User Interaction: fashion designers are able to browse through the template library, search for a specific template and choose including style or the season or the type of material it is made from for instance.
3. Expected Output: The system shows the selected template, ready for further customization and editing.

2.2.2 Module 2: Sketch to Image Conversion

Following are the requirements for module 2:

1. Functionality: Converts hand-drawn sketches into detailed 2D digital images using a fine-tuned Stable Diffusion model.
2. User Interaction: Designers upload sketches and provide relevant descriptions or tags; the system processes these inputs.
3. Expected Output: High-resolution digital images that closely resemble the original sketches are produced.

2.2.3 Module 3: 3D Modeling and Visualization

Following are the requirements for module 3:

1. Functionality: Transforms 2D digital images into 3D models that are rotatable and viewable from any angle.

2. User Interaction: Once a 2D image is created, users can initiate 3D modeling with options to adjust lighting, angles, and zoom.

3. Expected Output: A fully interactive 3D model of the clothing item is available for detailed inspection and further modifications.

2.2.4 Module 4: Virtual Try-On

Following are the requirements for module 4:

1. Functionality: Simulates how clothing items look on virtual avatars using the VITON model.

2. User Interaction: Users select a garment, choose an avatar or input body measurements, and then view the garment on the avatar.

3. Expected Output: A visualization of the garment as worn by the avatar, providing realistic representations of fit and style.

2.2.5 Module 5: Crossover Design Mechanism

Following are the requirements for module 5:

1. Functionality: Allows for the fusion of elements from two different garments to create a new, unique design.

2. User Interaction: Designers choose two existing garments or design elements, and specify how they should be combined.

3. Expected Output: A new garment design that incorporates the selected elements from the original garments.

2.2.6 Module 6: Clothing Image Ads

Following are the requirements for module 6:

1. Functionality: Automatically generates multiple marketing images featuring models wearing the designed clothes in various settings.

2. User Interaction: Users select the garment, choose the model poses, and the background settings.

3. Expected Output: Professionally styled images suitable for marketing purposes, displaying the clothes in diverse, appealing poses.

2.2.7 Module 7: Video Ads

Following are the requirements for module 7:

1. Functionality: Creates short video clips showcasing the clothes in dynamic custom scenes.
2. User Interaction: Users select clothes, define scene settings, and choose the sequence of model actions.
3. Expected Output: A 5-10 second marketing video that highlights the garment features dynamically, suitable for social media and advertising campaigns.

2.3 Non-Functional Requirements

2.3.1 Reliability

1. Mean Time Between Failures (MTBF): The system shall have an MTBF of no less than 100 hours, ensuring that software failures are infrequent during normal operation.
2. Protection from Failure: The system will implement regular data backups, automated recovery processes, and fail-over mechanisms to ensure minimal loss of progress in the event of a failure.

2.3.2 Usability

1. Ease of Use: The system shall allow users to complete common design tasks, such as selecting a template or uploading a sketch.
2. Accessibility: The system shall comply with web accessibility standards, ensuring that it is fully navigable for users.

2.3.3 Performance

1. Response Time: The system shall generate high-resolution digital images from user-uploaded sketches, regardless of the complexity of the sketch. 3D models from 2D digital images shall be rendered in within a minute.Virtual try-on simulations should generate results within 50 seconds of input. Marketing Content Generation module will generate a content within 3-4 minutes.

2. System Load: The system must handle up to 300 concurrent users without noticeable degradation in performance.
3. Data Processing: The system must process and store data efficiently, ensuring that uploading and processing of images under 50MB happen in less than a minute.

2.3.4 Security

1. Protection from Unauthorized Access: The system must implement robust authentication and authorization mechanisms to ensure only registered users can access the design studio.

Table 2.2: External and Non-Functional Requirements

Requirement Category	**Requirement**	**Description**
NFR	Reliability	The system shall have an MTBF of no less than 100 hours, ensuring that software failures are infrequent during normal operation.
NFR	Reliability	The system will implement regular data backups, automated recovery processes, and fail-over mechanisms to ensure minimal loss of progress in the event of a failure.
NFR	Usability	The system shall allow users to complete common design tasks, such as selecting a template or uploading a sketch.The system shall comply with web accessibility standards, ensuring that it is fully navigable for users.
NFR	Performance	The system shall generate high-resolution digital images from user-uploaded sketches within a minute and virtual try-on simulations within 50 seconds of input. Marketing Content Generation module will generate content within 3-4 minutes.The system must handle up to 300 concurrent users without noticeable degradation in performance.The system must process and store data efficiently, ensuring that uploading and processing of images under 50MB happen in less than a minute.
NFR	Security	The system must implement robust authentication and authorization mechanisms to ensure only registered users can access the design studio.
NFR	Compliance	The system shall comply with GDPR, CCPA, and other relevant data protection regulations to ensure that user data is handled securely and lawfully. Compliance with copyright and intellectual property laws must be ensured to protect the original works of users and prevent any unauthorized use of copyrighted materials within the system.
External Requirements	External System Integration	The system shall integrate with major cloud service providers (e.g., AWS, Azure) to utilize their computational resources and storage solutions.

2.4 Domain Model

A domain model for the AI Fashion Design Studio captures the essential entities, their characteristics, and interactions that enable fashion design and marketing. It includes entities such as "User," "Hand-drawn Sketch," "Template," and "Design," as well as processes like "Sketch to Image Conversion," "Virtual Try-On," and "3D Visualization." These components are interconnected to allow users to upload sketches and images, select templates, and seamlessly transition through design creation, virtual fitting, and the generation of marketing materials. The model facilitates the conversion of initial designs into ready-to-market digital formats, leveraging AI to enhance creativity and efficiency in fashion design.

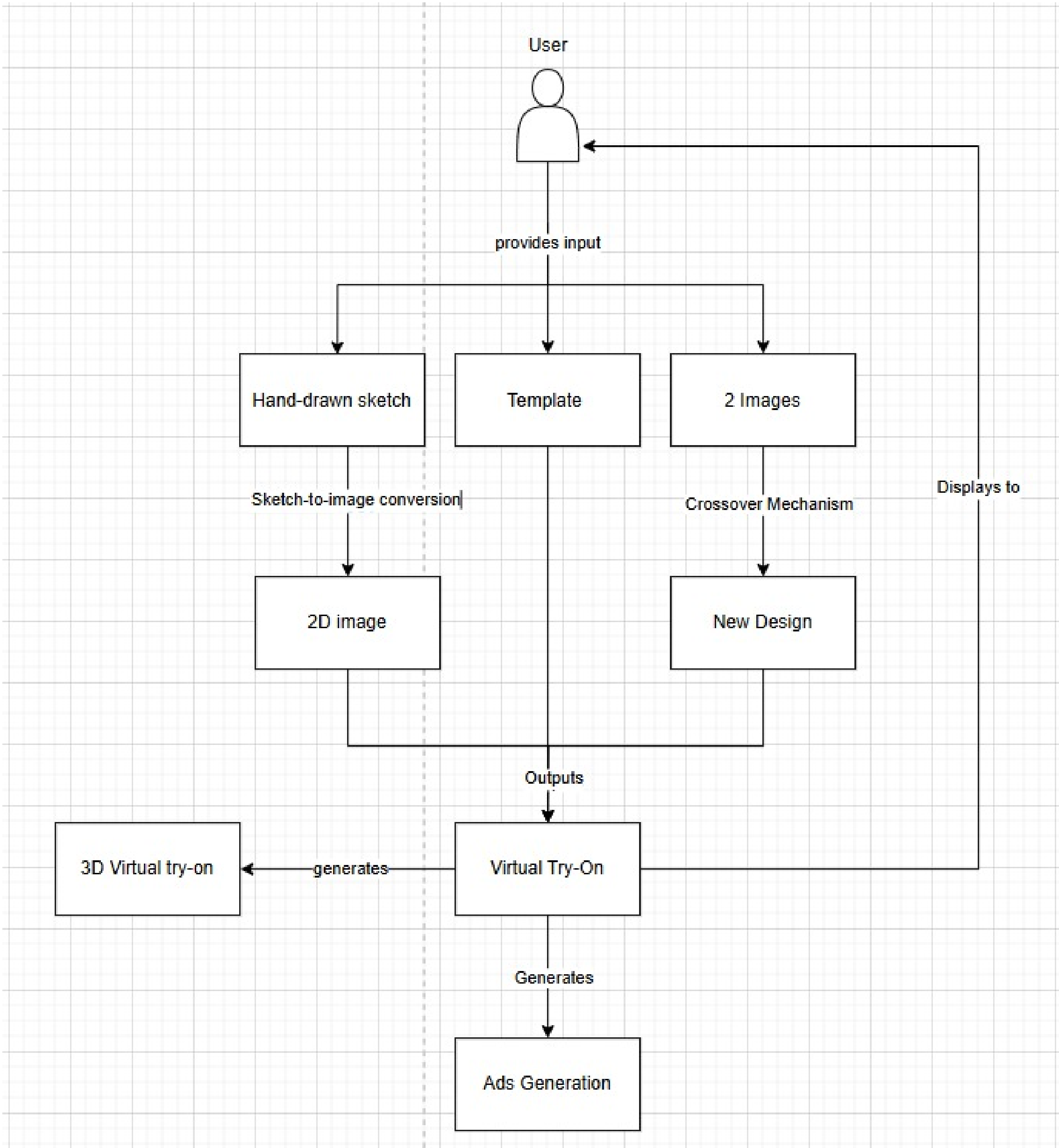

Figure 2.2: Domain Model

Chapter 3

System Overview

The AI Fashion Design Studio leverages generative AI technologies for fashion design creation, visualization and marketing. It offers a wide range of modules such as, Template Management module where the user is able to obtain a list of existing readily designed templates, A GUI which involves the easy interaction between the front end and back end processes, Sketch to Image Conversion where sketches are easily converted to digital images. Also, the platform includes a Virtual Try-On and 3D Visualization module where designers can re-imagine clothing designs on avatars, and a Crossover Design Mechanism for generating unique garments by combining elements from two existing designs.The tool also automatically generates marketing materials, leveraging diffusion models to create visually appealing image and video ads.This platform not only streamlines the design process but also enables rapid experimentation and customization, setting a new standard in digital fashion tools.

3.1 Architectural Design

An architectural diagram visually represents the structure and behavior of a system, showing its different components and how they interact. Our multi-layer architecture diagram details how user interactions, data processing, and storage are structured across different layers of the AI Fashion Design Studio. At the top, the User Interface layer captures user inputs like uploading sketches or selecting templates. The middle, Application Layer, processes these inputs through various mechanisms such as sketch to image conversion, virtual try-on, and ad generation. Finally, the Data Storage Layer manages the storage of all templates, images, videos, and other outputs, ensuring that all data is systematically organized and retrievable for future use and marketing efforts.

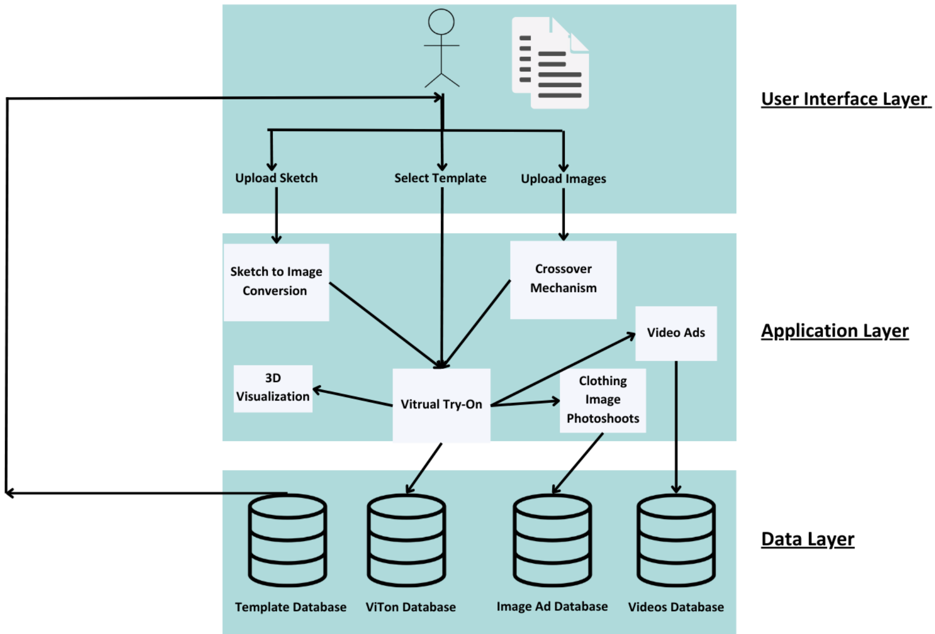

Figure 3.1: Architectural Diagram

3.2 Design Models

3.2.1 Activity Diagram

This activity diagram represents the workflow of the AI Fashion Design Studio, it starts with user providing input through three different paths: uploading a sketch, choosing from pre-designed templates, or uploading two images of clothes for the crossover mechanism. The sketch is converted to 2D image, which, along with the new design from the crossover mechanism, converge at the virtual try-on stage. This pivotal step allows users to visualize clothing on avatars. From the virtual try-on stage, the process diverges into three branches according to user choice: 3D visualization, clothing image ad generation, and video ad generation, each of which produces specific marketing and visualization outputs tailored to showcase the designs in dynamic ways.

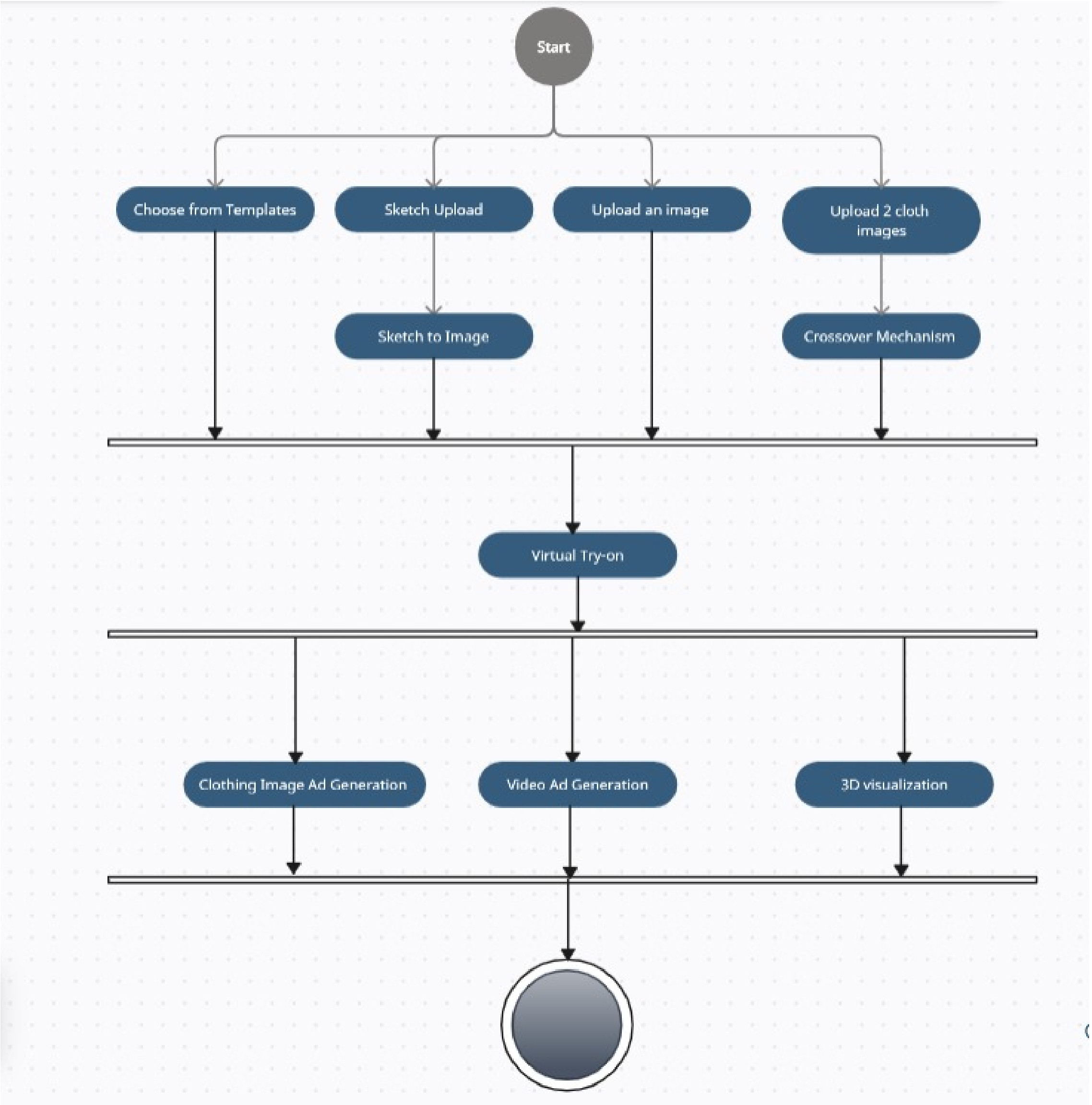

Figure 3.2: Activity Diagram

3.2.2 System Sequence Diagram

The system sequence diagram for the AI Fashion Design Studio illustrates the sequence of interactions between the user and the system across multiple modules. It begins with the user's input, which can be an uploaded sketch, a selected template, or multiple images for a crossover mechanism. These inputs are processed through distinct pathways: sketches are converted to 2D images, templates initiate virtual try-on, and multiple images lead to new garment designs via the crossover mechanism. All paths converge at the virtual try-on stage. Successful outputs from this stage flow into 3D visualization, and direct generation of clothing image ads and video ads. This structure ensures a clear and efficient progression from user input to the final visual output.

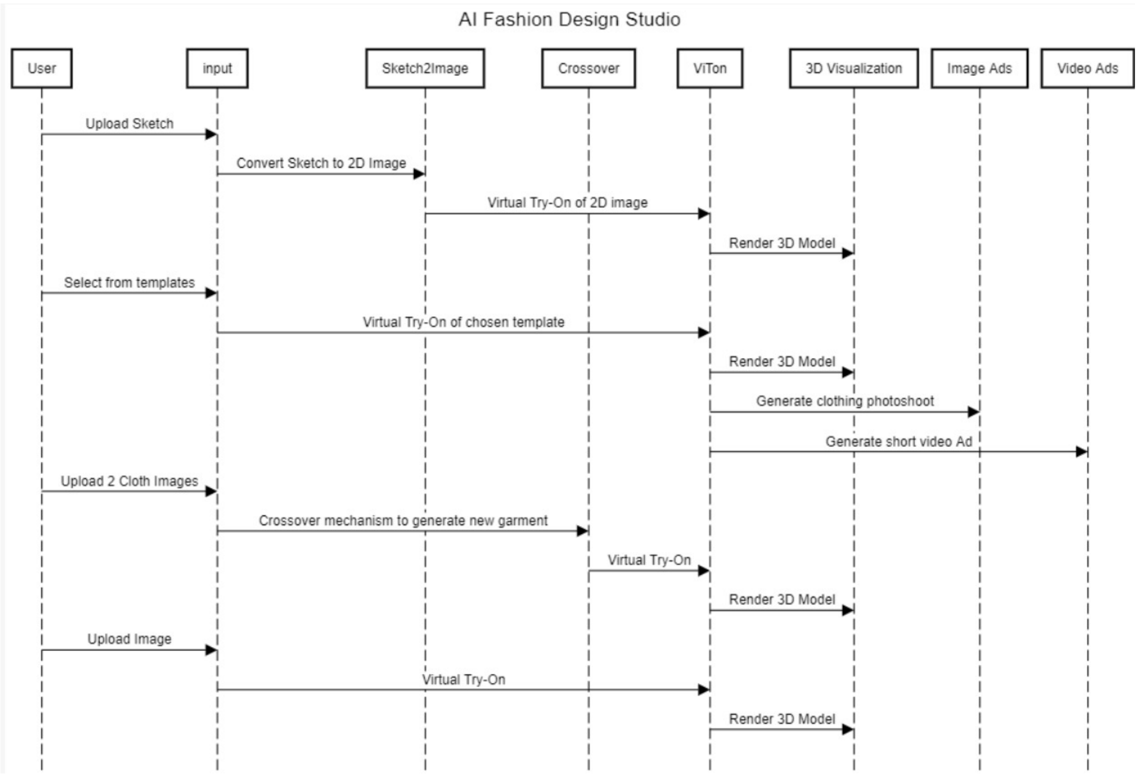

Figure 3.3: System Sequence Diagram

3.2.3 State Transition Diagram

The state transition diagram for AI Fashion Design Studio maps out the user's interaction path from initial input to final output across various design stages. It begins with three distinct starting points: uploading a sketch, selecting from pre-designed templates, or uploading multiple images for the crossover mechanism. Each path converges at the 'Sketch to Image' conversion, where sketches are transformed into 2D images, or the

'Cross-Over Mechanism', creating new designs. These elements then feed into a central 'Virtual Try-On' process, which visualizes the designs on a digital avatar. From this pivotal point, the workflow diverges into three separate outputs: generating image ads, creating video ads, and producing 3D visualizations, each branching out to deliver specific visual content and promotional materials directly related to the initial user inputs.

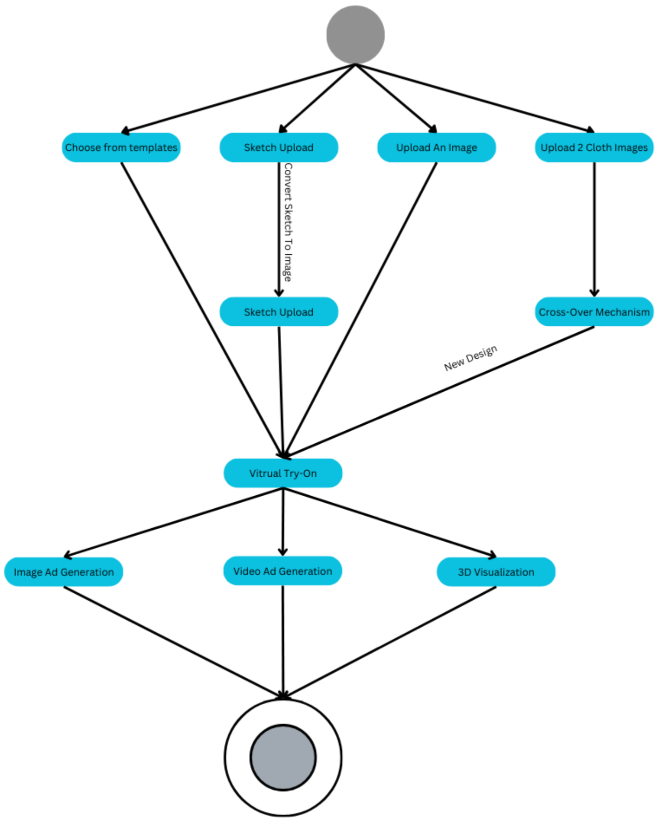

Figure 3.4: State Transition Diagram

3.2.4 Data Flow Diagram Level(0)

The AI Fashion Design Studio data flow diagram illustrates the primary interactions between customers, the studio, and marketing teams. Customers upload sketches or model images, which the studio converts into 3D models or virtual try-on, storing all data in a central database. This processed data is then utilized by marketing teams to create promotional content, enhancing outreach and customer engagement

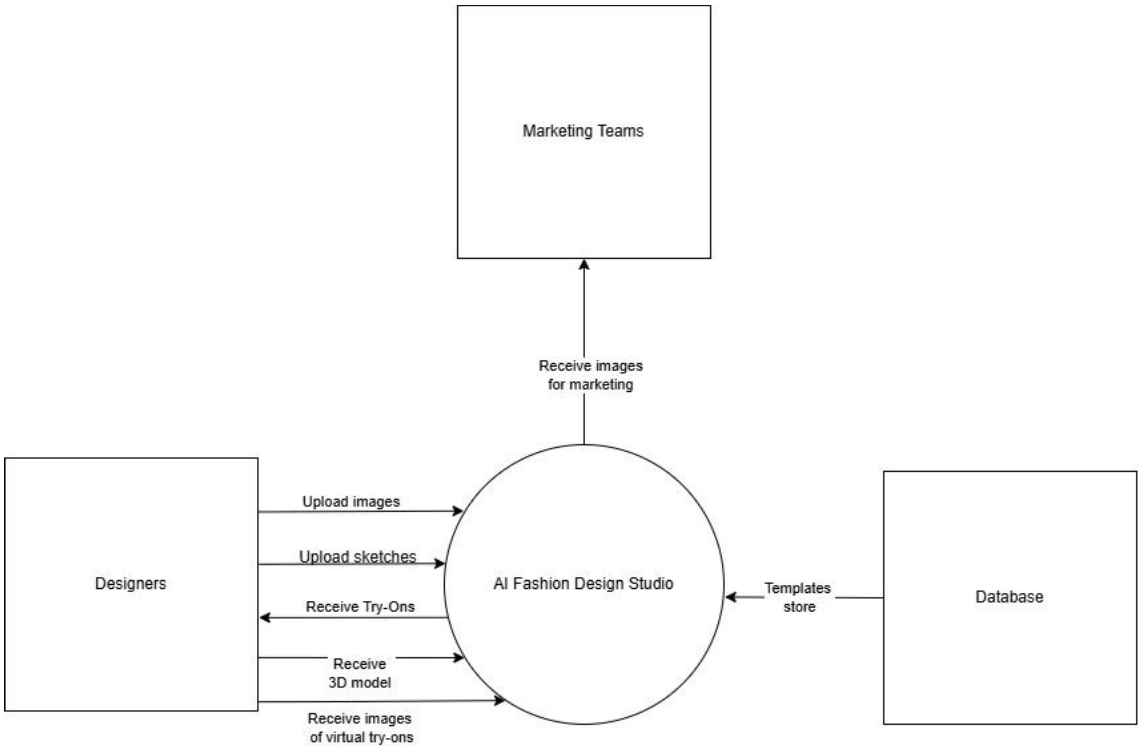

Figure 3.5: Data Flow Diagram Level(0)

3.2.5 Data Flow Diagram Level(1)

The Level 1 Data Flow Diagram for the AI Fashion Design Studio details the user's interaction with the system, where users can upload sketches for conversion, select and use templates, or upload model images for virtual try-ons. The system processes these inputs through the Sketch to Image Conversion, Template Management, and Virtual Try-On modules before generating marketing materials like video ads and clothing image ads. The outputs are then stored in the Advertisement Database and used by the Marketing Team for promotional activities, all managed through the centralized User Info and Template Databases.

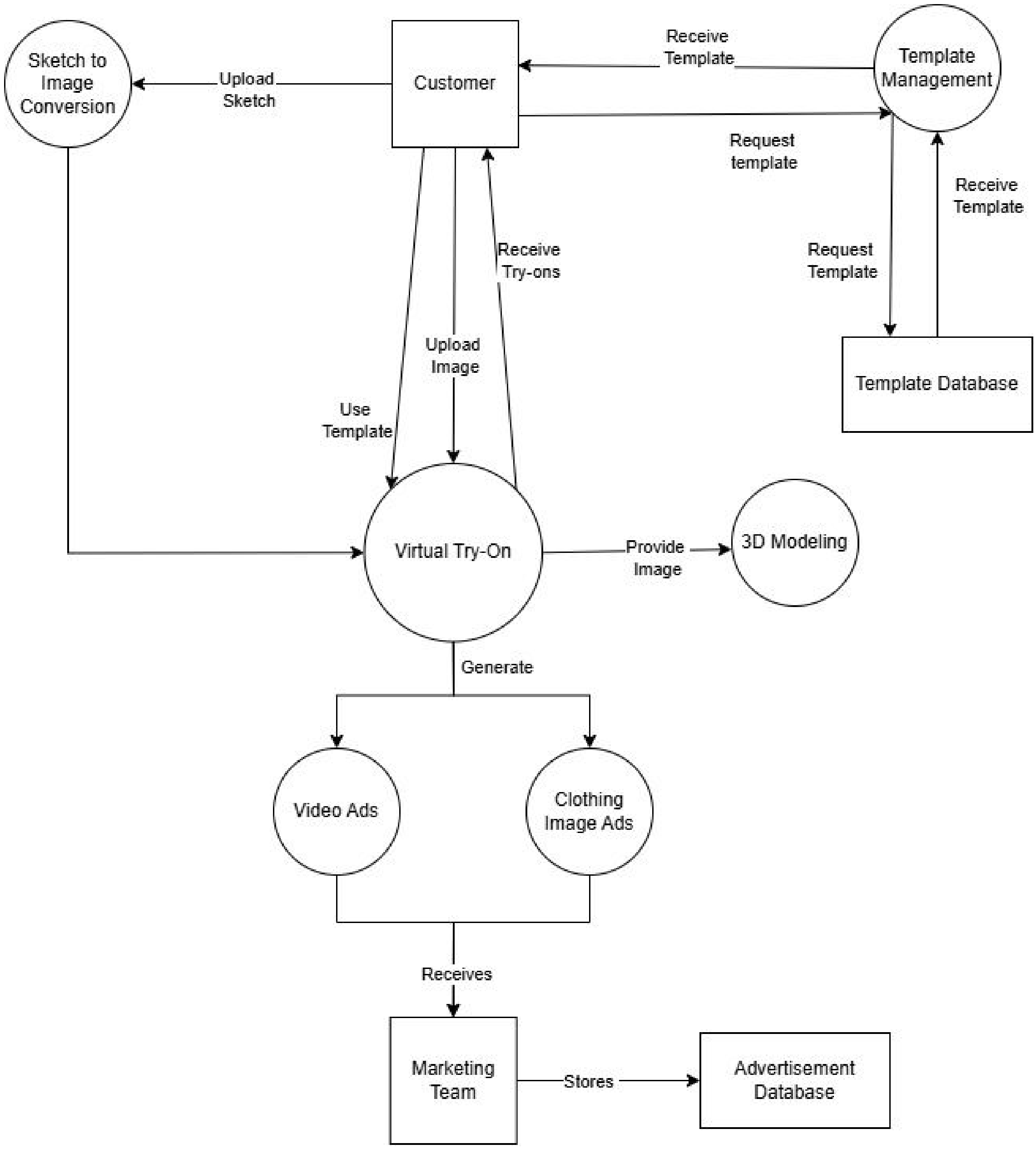

Figure 3.6: Data Flow Diagram Level(1)

3.3 Data Design

The data design for the AI Fashion Design Studio outlines how data flows through the system, detailing how it is processed, stored, and organized. The key entities include user inputs such as sketches and images, converted digital assets, virtual try-on, and marketing materials.

3.3.1 Major Data Entities and Structures

1. **User Inputs:**

   (a) Data Structure: Inputs include sketches, images of clothes, or selections from templates. These are handled as image files or selections from a predefined database.

   (b) Storage: Uploaded sketches and images are temporarily stored on secure cloud storage during processing.

   (c) Processing: Inputs are processed to convert sketches to 2D images or selected templates to initiate virtual try-on.

2. **2D Images and Designs:**

   (a) Data Structure: Images are processed into 2D digital designs, stored as high-resolution image files.

   (b) Storage: These images are stored in a dedicated directory within the system until they are either discarded after use or downloaded by the user.

   (c) Processing: Images undergo enhancements and adjustments in design parameters like color and style before entering the virtual try-on phase.

3. **Virtual Try-On Results:**

   (a) Data Structure: The output from virtual try-on includes images showing garments on avatars, stored as image files.

   (b) Storage: Results are temporarily held in a runtime environment to allow quick adjustments and previews.

   (c) Processing: These images are used to provide real-time visual feedback to users and for further refinement in 3D visualization.

4. **3D Models:**

   (a) Data Structure: Digital garments and avatars are rendered in 3D, managed as 3D model files.

   (b) Storage: 3D models are cached during sessions to facilitate swift interactions and adjustments.

   (c) Processing: Models are rendered from 2D designs, allowing users to inspect garments from multiple angles.

5. **Marketing Materials (Images and Videos):**

(a) Data Structure: Advertising content such as image ads and video ads generated from the design data, stored as multimedia files.

(b) Storage: Marketing materials are saved within the project's marketing content directory on the server.

(c) Processing: Content is generated using templates and customized settings to reflect the final product appearance in marketing campaigns.

6. **User Feedback and Adjustments:**

   (a) Data Structure: Feedback from users regarding designs, stored as text data or form inputs.

   (b) Storage: Feedback data is stored temporarily to adjust designs accordingly or permanently as insights for future improvements.

   (c) Processing: Feedback is analyzed to make informed adjustments to designs or marketing strategies.

3.3.2 Data Flow and Processing Stages

1. **User Uploads Inputs:** Users start by uploading sketches or images. These files are stored temporarily on a secure server. The system then processes these files to convert sketches into digital images or selects templates for further actions.

2. **Image and Design Processing:** Once the initial images are uploaded, they are converted into 2D digital designs or processed through a crossover mechanism to create new garment designs. These new designs are stored and queued for virtual try-on.

3. **Virtual Try-On:** The 2D images are used in the virtual try-on module where they are digitally fitted onto avatars. This process provides users with a realistic depiction of how the clothes will look when worn. The results are temporarily stored for immediate review.

4. **3D Visualization:** Successful try-on are moved to 3D visualization, where they are rendered into detailed 3D models allowing users to view garments in a fully rotatable interface. This stage enhances the design's details and realism, crucial for final adjustments.

5. **Generation of Marketing Materials:** Concurrently, the design data flows into marketing material generation modules, where both image and video advertisements are created. These materials are crafted to showcase the garments in dynamic, realistic settings, emphasizing their features.

6. **Output Delivery:** Finally, the processed images, 3D models, and marketing videos are compiled and presented to the user for approval. This comprehensive display helps users make informed decisions about final edits or confirmations before the designs go to market.

3.3.3 Data Storage and Organization

1. **File System:** All uploaded sketches and images are stored in the `/uploads/` directory on the server. As they are converted into 2D images or processed into new designs, these files are moved to a secure `/processed/` directory.

2. **Generated Outputs:** The final outputs, including 2D converted sketches, 3D models, and marketing materials (image and video ads), are stored in structured formats. Images and videos are saved in respective media folders and linked within our database to maintain organization and ease of access.

These storage and organization methods ensure that data is handled efficiently throughout the design process, providing a seamless and secure experience from sketch upload to final marketing output in the AI Fashion Design Studio.

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